# Kristina Dahle (Wilson)

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## EDUCATION

**Vanderbilt University.** Nashville, TN **Expected May 2029**
Ph.D. Mathematics

**University of Tennessee.** Knoxville, TN **May 2024**
B.S. Honors Mathematics **GPA 3.99**
*Minor*. Computer Science

**McMinn County High School.** Athens, TN **May 2020**

## PROFESSIONAL EXPERIENCE

**Research Assistant.** Mentors: Kwai Wong, Stanimire Tomov **May 2023 – July 2024**

- Developed C++ routines in Magma, a collection of matrix algebra libraries on GPU and multicore architectures.
- Designed and optimized routines to perform multidimensional tensor multiplication.
- Built routines to perform multidimensional tensor transposition.

**Research Assistant.** Mentor: Ioannis Sgouralis **Dec 2022 – Sep 2023**

- Derived ordinary differential equations to describe an NMR experiment.
- Developed software in MATLAB to simulate a sequence of 2D-NMR experiments.

**Research Assistant.** Mentor: Olivia Prosper **Feb 2022 – May 2023**

- Implement and study mathematical models in Python to analyze the spread of infectious disease.
- Modeled difference equations in Python and Excel to study traffic conditions.

## PUBLICATIONS

"Implementing Single-precision and Half-precision Tensor Operations," with C. Li and H.M. Lau, *Practice and Experience in Advanced Research Computing,* 2024.

"Modeling Traffic Conditions to Determine Shortest Path" with N. Dahle, *Society for Industrial and Applied Mathematics Undergraduate Research* (SIURO). 2023.

## PRESENTATIONS

"Implementing Tensor Operations in Magma," Practice & Experience in Advanced Research Computing Conference in Providence, Rhode Island. July 2024.

"Implementing 3-dimensional QR codes," with N. Dahle, at the University of Tennessee Knoxville. May 2024.

“Computational modeling in 2D NMR spectroscopy,” Discovery Day at the University of Tennessee Knoxville. September 2023.

“Implementing MAGMA tensor interfaces” with C. Li and E. Villasana, Summer Research Scholars Symposium at the University of Tennessee Knoxville. July 2023.

## HONORS AND AWARDS

**John H. Barrett Prize.** Awarded by the University of Tennessee Mathematics Department to the top 2 undergraduate seniors in the department. April 2024.

**Mary K Cook Memorial Mathematics Scholarship**. Awarded by the University of Tennessee Mathematics Department. August 2023.

**Faculty and Staff Honors Scholarship**. Awarded by the University of Tennessee Mathematics Department. April 2023.

**Dr. Warren L. & Mrs. Kyleen D. Franz Scholarship**. Awarded by the University of Tennessee Mathematics Department. April 2023.

**April Marie Morton Memorial Scholarship**. Awarded by the University of Tennessee Mathematics Department. April 2022.

**Trjitzsinksy Memorial Award Scholarship**. Awarded by the University of Tennessee Mathematics Department. August 2021.